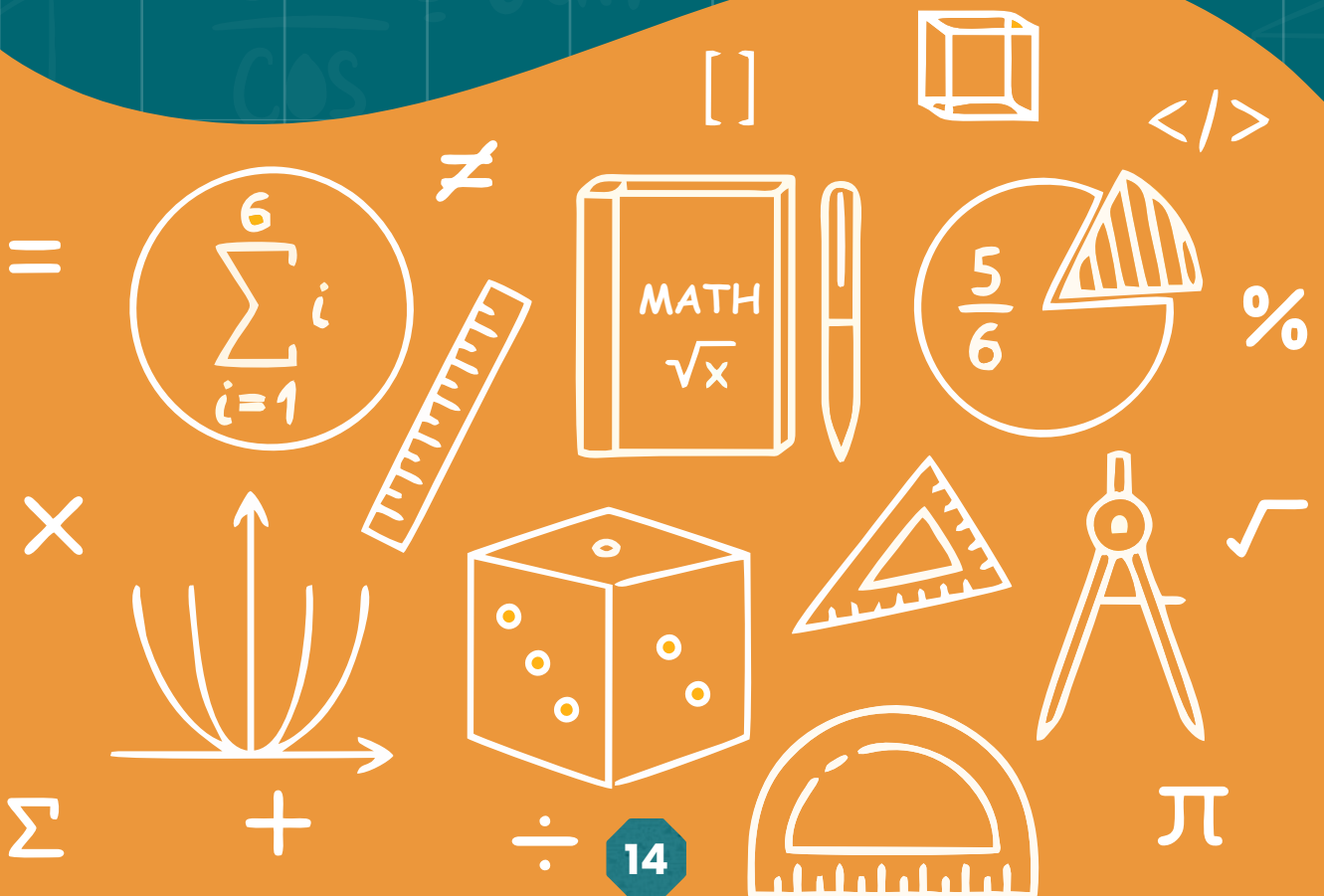


UNIT 2

REAL NUMBERS



RATIONAL NUMBERS

The numbers $\frac{2}{3}$, $\frac{1}{2}$, $\frac{5}{6}$ and $\frac{22}{7}$ represent the rational numbers. These numbers can be represented in the form of $\frac{p}{q}$ where p and q are integers and $q \neq 0$. The set of rational numbers is denoted by Q .

IRRATIONAL NUMBERS

Irrational numbers cannot be expressed in the form of a ratio of integers as $\frac{p}{q}$, where $p, q \in \mathbb{Z}$ and $q \neq 0$. If we take the square root of a number which is not a perfect square, it will be called an irrational number.

Example: π and $\sqrt{3}$ are examples of irrational numbers. The set of irrational numbers is denoted by Q' .

REAL NUMBERS

We use numbers in our daily life. Almost every number of this world we can think of is a real number.

For example, 0, -1, 12.59, -0.3846, 4, 30 and $\frac{3}{4}$ are the real numbers. Real numbers are a combination of rational and irrational numbers. Irrational numbers can be expressed in the form of decimal numbers which are non-terminating and non-recurring.

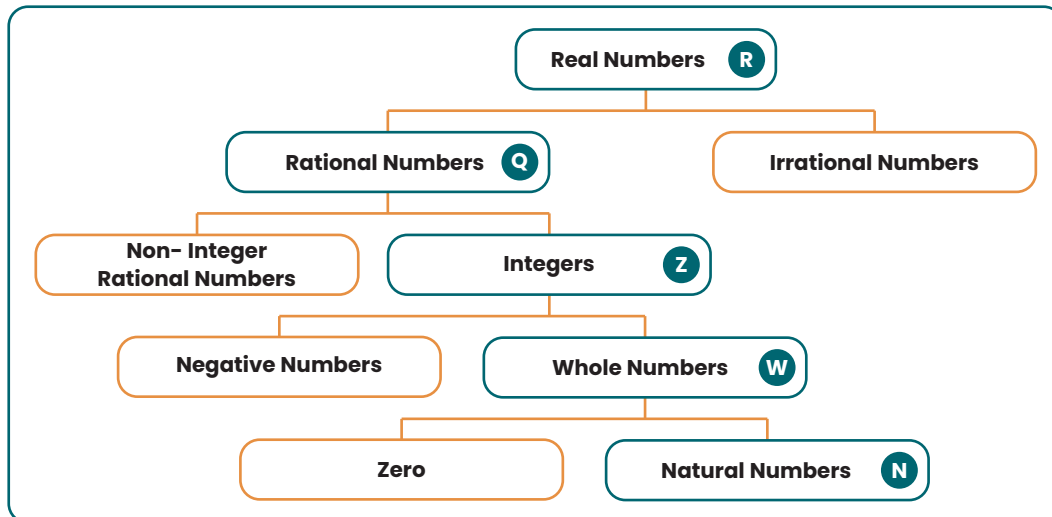
'Irrational numbers are non-terminating, non-repeating decimals.'

Recognize rational and irrational numbers:

Let us recognize rational and irrational numbers with the help of examples: The numbers $\frac{2}{8}$, 4.755, 5.0, 4, 9, 0.333..., -49 and $3\frac{3}{2}$ are **rational numbers**. The numbers $\sqrt{2}$, π , $\sqrt[3]{7}$, $\sqrt{48}$, and $\sqrt{122}$ are all **irrational numbers**. $\sqrt{2/5}$ and $1/3$ are irrational because in both, their numerator or denominator are not all integers.

Real numbers

A real number is a value that shows any quantity mapped on a number line. Union of the sets of rational numbers (Q) and irrational numbers (Q') is called the set of real numbers, denoted by R , such that $R = Q \cup Q'$. We can conclude it by this diagram:



Demonstrate non-terminating/non-repeating (or non-periodic) decimals

Let us consider the following examples:

Example 1: Convert $1/3$ into decimal.

Solution:

So, $1/3 = 0.333...$ or $0.\overline{3}$ is a non-terminating and repeating but periodic decimal (as 3 is repeating infinitely).

Note: If the remainder $\neq 0$, the number is called a non-terminating decimal.

$$\begin{array}{r}
 0.333... \\
 3 \overline{) 10} \\
 \underline{- 9} \\
 10 \\
 \underline{- 9} \\
 10 \\
 \underline{- 9} \\
 1
 \end{array}$$

Example 2: Convert $-4/7$ into decimal.

Solution:

So, $-4/7 = -0.571428...$

Example 3: Convert $1/2$ and $-4/5$ into decimal.

Solution:

(i) $1/2 = 0.5$, it is a terminating decimal.

(ii) $-4/5 = -0.8$, it is also a terminating decimal.

$$\begin{array}{r}
 -0.571428 \\
 7 \overline{) -40} \\
 \underline{+ 35} \\
 - 50 \\
 \underline{+ 49} \\
 - 10 \\
 \underline{+ 7} \\
 - 30 \\
 \underline{+ 28} \\
 - 20 \\
 \underline{+ 14} \\
 - 60 \\
 \underline{+ 54} \\
 - 4
 \end{array}$$

Note: If remainder = 0, the number is called a terminating decimal. Hence, every fraction is either terminating or non-terminating but a repeating decimal. So every fraction is a rational number. Irrational are numbers, whose digits in the decimal part are not terminating and not repeating in the same order, like $\sqrt{2} = 1.4142135...$, $\sqrt{5} = 2.23606...$, $\sqrt{243} = 15.5884...$ and $\pi = 3.1417...$

These are called irrational numbers.

EXERCISE 2.1

A. Separate rational and irrational numbers from the following:

- | | | | |
|-----------|----------------|-------------|----------------|
| 1) 35 | 2) $\sqrt{36}$ | 3) $-21/40$ | 4) $2\sqrt{4}$ |
| 5) 6π | 6) $22/7$ | 7) $2/5$ | 8) $\sqrt{7}$ |
| | | | 9) $\pi/5$ |

B. Identify terminating and non-terminating (periodic) fractions from the following rational numbers:

- | | | | |
|----------|-----------|-----------|-----------|
| 1) $1/7$ | 2) $2/3$ | 3) 0 | 4) $1/11$ |
| 5) $4/3$ | 6) $-1/5$ | 7) $-5/6$ | 8) 2256 |

SQUARES

The product of a number multiplied by itself is called its square.

For example, square of 2 is $2^2 = 2 \times 2 = 4$. In the same way, square of a is $a^2 = a \times a$.

PERFECT SQUARE

A natural number which is square of another natural number is called a perfect square.

For example, $9 = 3^2$. So, 9 is a perfect square of 3. Similarly, $16 = 4^2$ So, 16 is a perfect square of 4.

Example 1. Find the perfect squares of 11 and 65.

Solution:

Perfect square of 11 = $11^2 = 11 \times 11 = 121$.

Perfect square of 65 = $65^2 = 65 \times 65 = 4225$.

Establish patterns for the squares of natural numbers

Let us establish a pattern of adding natural numbers to get squares of natural numbers.

The first natural number is 1, therefore $1^2 = 1 \times 1 = 1$.

Now $2^2 = 2 \times 2 = 4 = 2 + 1 + 1$ or $2^2 = 1 + 2 + 1$ (by adding current number to be squared with twice the numbers before it).

In the same way: $3^2 = 3 \times 3 = 9 = 1 + 2 + 3 + 2 + 1$

Hence the desired pattern of squares by addition is as under:

$1^2 =$	1	$= 1$
$2^2 =$	$1 + 2 + 1$	$= 4$
$3^2 =$	$1 + 2 + 3 + 2 + 1$	$= 9$
$4^2 =$	$1 + 2 + 3 + 4 + 3 + 2 + 1$	$= 16$
$5^2 =$	$1 + 2 + 3 + 4 + 5 + 4 + 3 + 2 + 1$	$= 25$
$6^2 =$	$1 + 2 + 3 + 4 + 5 + 6 + 5 + 4 + 3 + 2 + 1$	$= 36$
$7^2 =$	$1 + 2 + 3 + 4 + 5 + 6 + 7 + 6 + 5 + 4 + 3 + 2 + 1$	$= 49$
$8^2 =$	$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1$	$= 64$
$9^2 =$	$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1$	$= 81$
$10^2 =$	$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1$	$= 100$

EXERCISE 2.2

A. Find perfect squares of the following:

- | | | | |
|-------|-------|-------|--------|
| 1. 5 | 2. 13 | 3. 19 | 4. 39 |
| 5. 45 | 6. 56 | 7. 63 | 8. 108 |

B. Find all the perfect squares between:

- | | |
|--------------|--------------|
| 1. 10 and 30 | 2. 20 and 50 |
| 3. 40 and 70 | 4. 70 and 90 |

C. Write patterns of additions for the following squares:

- | | | | |
|----------|-----------|-----------|-----------|
| 1. 7^2 | 2. 9^2 | 3. 15^2 | 4. 11^2 |
| 5. 8^2 | 6. 12^2 | 7. 16^2 | 8. 13^2 |

IRRATIONAL NUMBERS

The square root is an operation opposite to finding the perfect square of a number. For example, the perfect square of 4 is 16 as $4^2 = 16$. Therefore, the square root of 16 is 4.

We use the symbol of $\sqrt{\quad}$ for square root. So that $\sqrt{\quad}$ multiplied by itself, gives the original number.

Square root of a natural number

Example 1. Find the square root of the following by prime factorization and division methods:

- (i) 625. (ii) 1600.

Solution:

- (i) Square root of 625

Prime factorization method:

$$625 = 5 \times 5 \times 5 \times 5$$

or

$$\begin{aligned}\sqrt{625} &= \sqrt{(5 \times 5 \times 5 \times 5)} \\ &= 5 \times 5 \\ &= 25\end{aligned}$$

5	625
5	125
5	25
5	5
	1

Division method:

	25
2	625
+2	4
45	225
+5	225
50	0

$$1 \times 1 = 1$$

$$2 \times 2 = 4$$

$$3 \times 3 = 9$$

$$4 \times 4 = 16$$

$$5 \times 5 = 25$$

$$6 \times 6 = 36$$

$$\text{So, } \sqrt{625} = 25$$

Solution:(ii) Square root of 1600
Prime factorization method:

$$\text{So, } 1600 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 5 \times 5$$

or

$$\begin{aligned}\sqrt{1600} &= \sqrt{(2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 5 \times 5)} \\ &= 2 \times 2 \times 2 \times 5 \\ &= 40\end{aligned}$$

2	1600
2	800
2	400
2	200
2	100
2	50
5	25
5	5
	1

Division method:

	40
4	1600
+4	16
8	00

$$2 \times 2 = 4$$

$$3 \times 3 = 9$$

$$4 \times 4 = 16$$

$$5 \times 5 = 25$$

Single $\sqrt{\quad}$ from the pair of terms is shifted to square root.
So, $\sqrt{1600} = 40$

The square root of a common fraction (when numerator and denominator are perfect squares)

Example 2. Find the square roots of the following by prime factorization and division methods:

(i) $9/16$

(ii) $49/64$

Solution:

(i) Square root of $9/16$

Prime factorization method:

Numerator: 9

$$9 = 3 \times 3$$

or

$$\sqrt{9} = \sqrt{(3 \times 3)} = 3$$

3	9
3	3
	1

Denominator: 16

$$\begin{aligned} 16 &= 2 \times 2 \times 2 \times 2 \\ \sqrt{16} &= \sqrt{(2 \times 2 \times 2 \times 2)} \\ &= 2 \times 2 \\ &= 4 \end{aligned}$$

2	16
2	8
2	4
2	2
	1

Division method:

Numerator: 9

	3	
3	9	
+3	9	
6	0	

$$1 \times 1 = 1$$

$$2 \times 2 = 4$$

$$3 \times 3 = 9$$

So, $\sqrt{9} = 3$

Denominator: 16

	4	
4	16	
+4	16	
8	0	

or $\sqrt{16} = 4$

Therefore:

$$\sqrt{\frac{9}{16}} = \frac{\sqrt{9}}{\sqrt{16}} = \frac{3}{4}$$

Solution:

(ii) Square root of $49/64$

Prime factorization method:

Numerator: 49

$$\begin{aligned} 49 &= 7 \times 7 \\ \sqrt{49} &= \sqrt{(7 \times 7)} \\ \sqrt{49} &= 7 \end{aligned}$$

7	49
7	7
	1

Denominator: 64

$$\begin{aligned} 64 &= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \\ \text{or} \\ \sqrt{64} &= \sqrt{(2 \times 2 \times 2 \times 2 \times 2 \times 2)} \\ &= 2 \times 2 \times 2 \\ &= 8 \end{aligned}$$

2	64
2	32
2	16
2	8
2	4
2	2
	1

Division Method:

Numerator: 49

	7	
7	49	
+7	49	
14	0	

$$\sqrt{49} = 7$$

Denominator: 64

	8	
8	64	
+8	64	
16	0	

$$6 \times 6 = 36$$

$$7 \times 7 = 49$$

$$8 \times 8 = 64$$

$$\text{Or } \sqrt{64} = 8$$

Therefore:

$$\sqrt{\frac{49}{64}} = \frac{\sqrt{49}}{\sqrt{64}} = \frac{7}{8}$$

The square root of a decimal (Perfect Square Form)

Example 3. Find the square root of the following by prime factorization and division methods:

(i) 0.01.

(ii) 1.21.

(iii) 0.64.

Solution: (i) To find the square root of 0.01

(a) Prime factorization method

$$0.01 = \frac{0.01}{1} = \frac{1}{100}$$

Numerator: 1, $\sqrt{1} = 1$

Denominator: 100

2	100
2	50
5	25
5	5
	1

$$\begin{aligned} 100 &= 2 \times 2 \times 5 \times 5 \\ \sqrt{100} &= \sqrt{(2 \times 2 \times 5 \times 5)} \\ &= 2 \times 5 \\ &= 10 \end{aligned}$$

Division Method:

	0.1
1	0.01
+1	1
2	0

So, $\sqrt{0.01} = 0.1$

Therefore:

$$\sqrt{0.01} = \sqrt{\frac{1}{100}} = \frac{\sqrt{1}}{\sqrt{100}} = \frac{1}{10} = 0.1$$

Solution: (ii) To find the square root of 1.21

(a) Prime factorization method and Division method:

Prime factorization method

$$0.21 = \frac{1.21}{100} = \frac{121}{100}$$

Numerator: 121

$$121 = 11 \times 11$$

or

$$\sqrt{121} = \sqrt{(11 \times 11)} = 11$$

Denominator: 100

$$100 = 2 \times 2 \times 5 \times 5$$

$$\sqrt{100} = \sqrt{(2 \times 2 \times 5 \times 5)}$$

$$= 2 \times 5$$

$$= 10$$

11	121
11	11
	1

2	100
2	50
5	25
5	5
	1

Therefore,

$$\sqrt{1.21} = \sqrt{\frac{121}{100}} = \frac{\sqrt{121}}{\sqrt{100}} = \frac{11}{10} = 1.1$$

Using long division method:

	1.1
1	1.21
+1	1
21	021
+1	21
22	0

So, $\sqrt{1.21} = 1.1$

EXERCISE 2.3

A. Find the square root of the following by prime factorization method.

1. 961

2. 1296

3. 344569

4. 817216

5. $81/121$

6. $4096/9604$

7. $1764/7744$

8. 249.64

B. Find the square root of the following by division method.

1. 1024

2. 14161

3. 996004

4. 1032796

5. $169/289$

6. $29241/55696$

7. 648.7209

8. 7613.609536

Find the square root of a number which is not a perfect square

All natural numbers and rational numbers are not perfect squares.

For example,

2, 3, 4, 6.4, 2.5135.

Division method is used to find the square root of such numbers up to required decimal places. These square roots are irrational numbers.

Example 1. Find the square root of 3 up to two decimal places.

Solution: As there is no decimal value in 3, we put two pairs of zeros to get the required decimal places.

	1.73
1	3.0000
+1	1
27	200
+7	189
343	1100
+3	1029
	71

$$26 \times 6 = 156$$

$$27 \times 7 = 189$$

$$28 \times 8 = 224$$

$$342 \times 2 = 684$$

$$343 \times 3 = 1029$$

$$344 \times 4 = 1376$$

Therefore, $\sqrt{3} = 1.73$

Example 2. Find the square root of 40.13 up to three decimal places.

Solution:

	6.334
6	40.130000
+6	36
123	413
+3	369
1263	4400
+3	3789
12664	67100
+4	50656
12668	10444

$$122 \times 2 = 244$$

$$123 \times 3 = 169$$

$$124 \times 4 = 496$$

$$1262 \times 2 = 2524$$

$$1263 \times 3 = 3789$$

$$1264 \times 4 = 5056$$

$$12663 \times 3 = 37989$$

$$12664 \times 4 = 50656$$

$$12665 \times 5 = 63325$$

Therefore, $\sqrt{40.13} \approx 6.334$
(up to 3 decimal places)

EXERCISE 2.4

A. Find the square root of the following up to 2 decimal places.

- | | | | |
|--------|-------|-------------|-----------|
| 1. 2 | 2. 5 | 3. 13 | 4. 4.141 |
| 5. 180 | 6. 25 | 7. 152.7696 | 8. 0.0143 |

B. Find the square root of the following up to 3 decimal places.

- | | | | |
|--------|--------|-----------|-----------|
| 1. 6 | 2. 8 | 3. 11 | 4. 45.125 |
| 5. 205 | 6. 7.3 | 7. 125.79 | 8. 1468.9 |

Use rule to determine the number of digits in the square root of a perfect square.

Number of digits in the square root of a perfect square can be calculated without finding the square root by the following rule:

Let n be the number of digits in a perfect square, then its square root contains:

- $n/2$ Digits when n is even and $n+1/2$ digits when n is odd

Example 1. Find the number of digits in the square root of 240100 and verify your answer.

Solution:

As the number of digits in 240100 is 6 which is an even number.

So, the number of digits in the square root of 240100 will be:

Verification:

		490
4		240100
+4	16	
89	801	
+9	801	
98	00	

Hence, $\sqrt{240100} = 490$ which has three digits.

EXERCISE 2.5

Q. Find the number of digits in the square roots of the following perfect squares:

- | | | | |
|-----------|-------------|-------------|---------------|
| 1. 8649 | 2. 250000 | 3. 322624 | 4. 614656 |
| 5. 432964 | 6. 81000001 | 7. 43388569 | 8. 3474748809 |

Solve real-life problems involving square roots

Example 1. In a room, 1089 marbles are arranged on the floor in such a way that the number of marbles in each row is the same as the number of rows. Find how many rows of marbles are there.

Solution:

Since the number of marbles in each row is the same as the number of rows, we have to find the square root of 1089.

	33	
3		1089
+3		9
63		189
+3		189
66		0

As $\sqrt{1089} = 33$

So, there are 33 rows of marbles.

Example 2. The area of a square field is 4225 square metres. Find the length of its side.

Solution:

As the area of a square = $(\text{side})^2$,

	65	
6		4225
+6		36
125		625
+5		625
130		0

So, Side = Area of square = $\sqrt{4225}$

As, $\sqrt{4225} = 65$

Therefore, the length of a side = 65 metres

Example 3. Find the least number:

(i) When subtracted from 124, (ii) When added to 124, (iii) When multiplied by 124, in order to make it a perfect square

Solution:

	11
1	124
+1	1
21	24
+1	21
22	3

Let us find the square root of the non-perfect square number 124.

As $124 = 11^2 + 3$ (greater than 11^2 by 3).

(i) Hence, the required least number is 3, which when subtracted from 124, i.e., $124 - 3 = 121$, gives a perfect square (less than 12^2 or 11^2 by 20).

(ii) As $124 = 12^2 - 20$, So 20 is the required least number which when added to 124, i.e., $124 + 20 = 144$, gives a perfect square (12^2).

Thus, $124 + 20 = 12 \times 12 = 144$ which is the perfect square.

2	124
2	62
31	31
	1

(iii) $124 = 2 \times 2 \times 31$. As 31 is not occurring in a pair, 31 is the required least number by which when 124 is multiplied, i.e., $2 \times 2 \times 31 \times 31$ gives a perfect square $(2 \times 31)^2$.

EXERCISE 2.6

1. A contribution was made for a party of students. Each student contributed as much as the number of students present. The collected amount was Rs 484. Find the amount paid by each.
2. The area of a square field is 10816 m^2 . Find the length of its side.
3. In a school assembly 1129 students were standing in rows such that the number of rows are equal to the number of students in a row. Find the number of students to leave to make it a complete square arrangement.
4. The area of a rectangular field is 1832 square metres. Its breadth is 44 metres. Find the length of the rectangular field.
5. The area of a square field is 3025.62 square metres. Find the length of wire required for fixing along the sides as a fence.

CUBES AND CUBE ROOTS

Recognize cubes and perfect cubes:

(i) Cube: The cube of a number is the number obtained by raising it to the power 3. It means the number is multiplied three times by itself.

Examples:

$$1 \times 1 \times 1 = 1^3 = 1$$

$$3 \times 3 \times 3 = 3^3 = 27$$

$$6 \times 6 \times 6 = 6^3 = 216$$

$$2 \times 2 \times 2 = 2^3 = 8,$$

$$5 \times 5 \times 5 = 5^3 = 125$$

This means: 1 is the cube of 1, 8 is the cube of 2, 27 is the cube of 3, 64 is the cube of 4, 125 is the cube of 5, and 216 is the cube of 6.

(ii) Perfect Cube: It is a number that is the result of multiplying an integer by itself three times, or it is an integer raised to the third power.

Examples:

$$7^3 = 7 \times 7 \times 7 = 343$$

$$9^3 = 9 \times 9 \times 9 = 729$$

$$8^3 = 8 \times 8 \times 8 = 512$$

$$10^3 = 10 \times 10 \times 10 = 1000$$

Hence 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000 are perfect cubes of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, respectively.

Similarly, we can find perfect cubes of decimals, like 1.1, 2.5, etc.

Perfect cube of 1.1 is $(1.1)^3 = (1.1) \times (1.1) \times (1.1) = 1.331$

Perfect cube of 2.5 is $(2.5)^3 = (2.5) \times (2.5) \times (2.5) = 15.625$.

Example 1: Examine whether these are perfect cubes or not:

(i) 1728

(ii) $5\frac{23}{64}$

(iii) 13.824

Solution: We can examine by making their prime factors. First of all, we find the prime factorization as under:

(i) $1728 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3$

$$= 1728 = 2^3 \times 2^3 \times 3^3 = (2 \times 2 \times 3)^3 = 12^3$$

Thus, 1728 is perfect cube of 12.

(ii) $5\frac{23}{64} = (5 \times 64 + 23)/64$

$$= (320 + 23)/64$$

$$= 343/64$$

$$\frac{343}{64} = \frac{7 \times 7 \times 7}{4 \times 4 \times 4}$$

$$= (7/4)^3$$

$$= \left(\frac{7}{4}\right)^3$$

Hence, $5\frac{23}{64}$ is perfect cube of $1\frac{3}{4}$

2	1728
2	864
2	432
2	216
2	108
2	54
3	27
3	9
3	3
	1

EXERCISE 2.7

A. Find the cubes of the following:

- | | | | |
|-------|-------|-------|-------|
| 1. 10 | 2. 20 | 3. 30 | 4. 40 |
| 5. 50 | 6. 60 | | |

B. Find the perfect cubes of the following fractions:

- | | | | |
|-------------------|-------------------|--------------------|--------------------|
| 1. $\frac{1}{2}$ | 2. $\frac{2}{3}$ | 3. $\frac{3}{4}$ | 4. $\frac{4}{5}$ |
| 5. $\frac{11}{3}$ | 6. $2\frac{1}{2}$ | 7. $11\frac{1}{4}$ | 8. $11\frac{1}{6}$ |

C. Find the perfect cubes of the decimals:

- | | | | |
|--------|--------|--------|--------|
| 1. 0.4 | 2. 0.9 | 3. 1.2 | 4. 1.6 |
| 5. 2.1 | 6. 2.5 | | |

D. Examine whether the following are perfect cubes or not:

- | | | | |
|---------------------|----------|----------|--------------------|
| 1. 512 | 2. 729 | 3. 3.375 | 4. $\frac{27}{64}$ |
| 5. $\frac{64}{125}$ | 6. 4.096 | | |

Find the cube roots of numbers which are perfect cubes.

We know that $8 = 2 \times 2 \times 2 = (2)^3$. It means the cube of 2 is 8, i.e., $2^3 = 8$, or we can say the cube root of 8 is 2.

It can be written as: $\sqrt[3]{8} = \sqrt[3]{(2 \times 2 \times 2)} = 2$ or $\sqrt[3]{8} = 2$.

Similarly, we can find the cube root of any perfect cube with the help of prime factorization.

Example: Find the cube root of:

1. $1000 = 10 \times 10 \times 10 = 10^3$
Therefore $1000 = 10^3$ or $(1000)^{1/3} = 10$

2. $216 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 = (2 \times 2 \times 2) \times (3 \times 3 \times 3)$
 $216 = 2^3 \times 3^3 = 6^3$ or $\sqrt[3]{216} = 6$

2	216	3	729
2	108	3	243
2	54	3	81
3	27	3	27
3	9	3	9
3	3	3	3
	1		1

Recognize Properties of Cubes of Numbers

Following are some recognized properties of cubes of numbers with examples:

(i) Cube of a positive number is always positive.

For example: $2^3 = 8$, $5^3 = 125$.

(ii) Cube of a negative number is always negative.

For example: $(-3)^3 = -27$, $(-6)^3 = -216$

(iii) Cube of an even number is always an even number.

For example: $2^3 = 8$, $4^3 = 64$

(iv) Cube of an odd number is always an odd number.

For example: $3^3 = 27$, $5^3 = 125$.

(v) Cubes satisfy the distributive properties as under:

(a) Multiplication and Division:

(i) $(2 \times 3)^3 = 2^3 \times 3^3$, $(8 \times 9)^3 = 8^3 \times 9^3$

(ii) $(4/5)^3 = 4^3/5^3$, $(6/7)^3 = 6^3/7^3$

(vi) Cube numbers are the perfect cubes.

For example: $7^3 = 343$, $9^3 = 729$

So, 343, 729 and 1728 are perfect cubes.

EXERCISE 2.8

A. Find the cube roots of the following:

1. 64

2. 343

3. 1,331

4. 512

5. 729

6. 6,172.8

7. 3,375

8. 13,824

B. Find the cube roots of the following fractions:

1. $1/8$

2. $2\frac{10}{27}$

3. $\frac{61}{64}$

4. $\frac{91}{125}$

5. $343/729$

B. Verify with examples that:

(i) The cube of a positive number is always a positive number.

(ii) The cube of a negative number is always a negative number.

(iii) The cube of every even number is always an even number.

(iv) The cube of every odd number is always an odd number.

C. Verify the following:

1. $(3 \times 4)^3 = 3^3 \times 4^3$

2. $(2/3)^3 = 2^3/3^3$

3. $(2 \times 5)^3 = 2^3 \times 5^3$

4. $(3/4)^3 = 3^3/4^3$

5. $(4 \times 5)^3 = 4^3 \times 5^3$

6. $(10 \times 20)^3 = 10^3 \times 20^3$

7. $(5/7)^3 = 5^3/7^3$